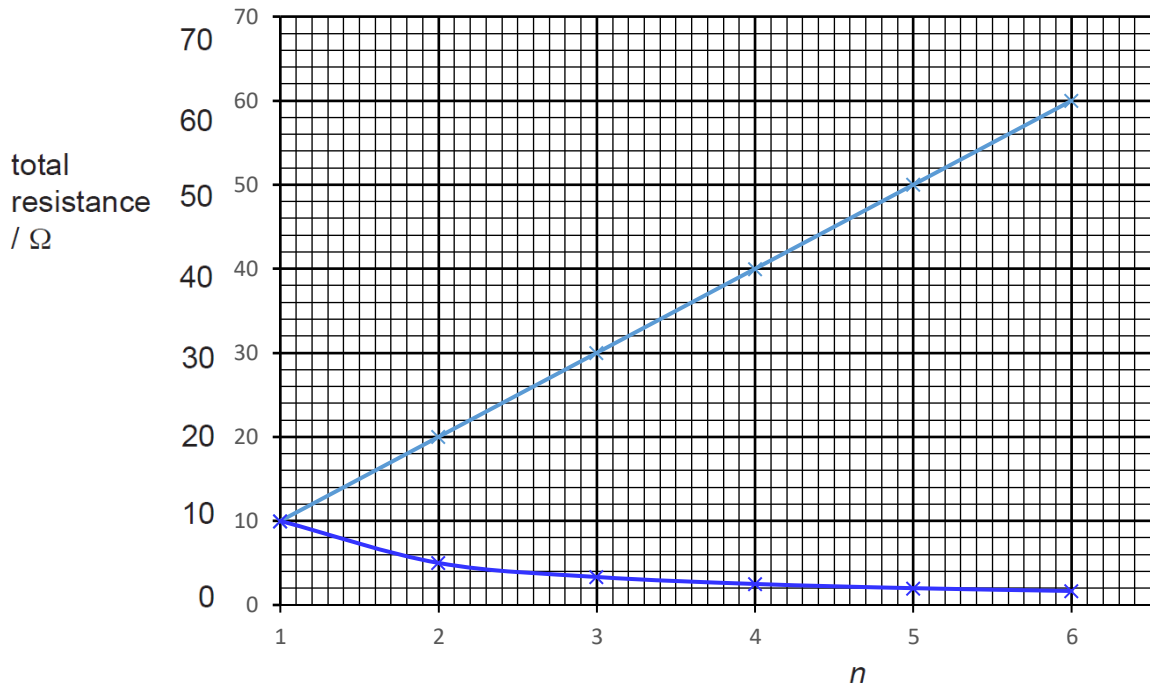


- 6 (a) (i) 1. Give an expression for the total resistance of n resistors, each of resistance $10\ \Omega$, connected in **series**, where n is the number of resistors.

resistance = $10n\ \Omega$ [1]

2. On Fig. 6.1, plot a graph showing the values of the total resistance for $n = 1, 2, 3, 4, 5$ and 6 . Label the line S.



[2]

Fig. 6.1

- (ii) 1. Give an expression for the total resistance of n resistors, each of resistance $10\ \Omega$, connected in **parallel**, where n is the number of resistors.

resistance = $10/n\ \Omega$ [1]

2. On Fig. 6.1, plot a graph showing the values of the total resistance for $n = 1, 2, 3, 4, 5$ and 6 . Label the line P. [2]

- (b) (i) Fig. 6.2 shows a network of 16 resistors, each of resistance $10\ \Omega$ connected in four parallel lines of four resistors in series.

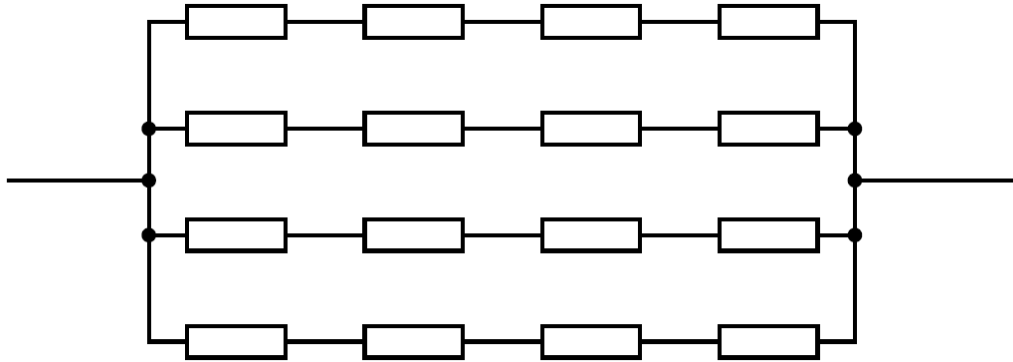


Fig. 6.2

Calculate the total resistance of the network.

$$R_{\text{tot}} = \frac{40}{4} = 10 \, \Omega$$

total resistance =10 [1]

- (ii) State the total resistance of a network of n parallel lines of n resistors in series where every resistor has a resistance of $10 \, \Omega$.

$$R_{\text{tot}} = \frac{10n}{n} = 10 \, \Omega$$

resistance = [1]

- (iii) Explain one practical advantage of using many individual resistors in this way, rather than using a single resistor.

If one resistor is faulty or inaccurate, the total resistance of the array is close to

 10 Ω , and so it's the array continues to function as it should.
 [1]

The current through each resistor in the array would be smaller than the current [Total 9]

through a single resistor. The resistors in the array would be less likely to reach